

HPAS Acceptance Testing Plan

Acceptance Test Plan — Hospital Patient Referral, Treatment and Administration System

Document Version	v1.0	Corresponding model	9 1 BPMN / 39 Camunda Forms / 17 flows Definition
System under test	HPAS on Camunda 8 (Spring Boot Job Worker)	Business Rule Baseline	BR-01 ~ BR-32、 OQ-01
Testing environment	Camunda 8 Run 8.10.0-alpha5 (self-built locally)	Performability	All test cases can be executed and verified in a local cluster.

1. Purpose & Scope

This plan establishes **clear and measurable** acceptance criteria for HPAS, and ensures that each criterion can be traced back to project requirements and **Business Rules**. The scope includes:

- End-to-end patient pathway:** Referral acceptance → Clinical decision-making → Appointment scheduling → Patient notification → Outpatient lettering → Funding determination → Payment processing → Follow-up consultation — comprising 8 business sub-processes (selecting 1–8), which are linked by Call activity in the overall flowchart to form a single main process.
 - Inter-institutional collaboration:** Each sub-process adopts a dual-Pool architecture consisting of an "Hospital Pool + External Pool," with messages exchanged via bpmn:messageFlow.
(External parties: GP/other hospitals, scheduling services, communication services/patients, funding agencies/insurance companies, PSP).
- Segregation of Duties and Compliance:** The three categories of duties—clinical, administrative, and financial—must not overlap; all decision-making, contact, payment, funding, and correspondence activities must be fully documented for audit purposes.**Abnormalities and Degradation:** missing items requiring supplementation, no available slots for scheduling, failed notifications, failed/duplicate payments, expired correspondence, and incorrect data entry.

Not covered by this scope: performance stress testing, multi-tenancy, SaaS authentication, and production-grade high availability; these will be addressed through separate non-functional testing plans.

2. General Principles of Acceptance Criteria

Number	Acceptance Criteria (Measurable)	Measurement Method
AC-1	Each choice (1–8) can be initiated by a non-technical user from the Tasklist and completed, with the process instance status set to COMPLETED.	Operate: instance status + no pending tasks
AC-2	In the overall flowchart, eight Call activities sequentially invoke their corresponding sub-processes; once a sub-process completes, the parent process continues, ultimately reaching the COMPLETED state.	Child instance parentProcessInstanceKey Correct; overall instance status
AC-3	Each business rule (BR) must be covered by at least one test case, and all covering test cases must pass.	Section 6: Coverage Matrix
AC-4	Each cross-Pool interaction generates a corresponding message stream and starts or completes an external party instance.	External Pool instances exist and are in a healthy state.
AC-5	For any use case, during execution incident = 0 (exception paths must be represented by BPMN branches and cannot be exposed via incidents)	Operate → Incident Count
AC-6	Key business operations are written to queryable variables (e.g., decision-maker, rationale, timestamp, transaction ID, etc.) to ensure audit trail compliance.	Variable Retrieval and Verification
AC-7	Models and forms can be deployed with a single click (submitting 48 files at once), with zero issues during structural validation.	deploy-all.ps1 + _validate.ps1

3. Preconditions

3.1 Environment Preconditioning (shared by All Test Cases)

- Camunda 8 has been started: c8run.exe start, REST http://127.0.0.1:8080, gRPC 127.0.0.1:26500, Log in demo/demo.
- Models and forms have been deployed: cd D:\test1;. \deploy-all.ps1 (9 BPMN models + 39 forms).
- Job Worker has run: cd D:\test1\java; mvn spring-boot:run (13 Worker classes / 31 topics, logs show Started WorkerApplication).
- The simulation switch is disabled by default (no SIMULATE_* parameters are set).

3.2 Test Data Baseline

Data item	Short-cut process	Use
Referral number	REF-<Scenario Number>(e.g., REF-TC03)	Referral verification / Clinical decision-making / Scheduling message association key
Reservation number	APPT-<Scenario ID>	Patient notification / funding approval message association key
Case Number	CASE-<Scenario Number>	Letter / Payment message association key
Appointment Date	Relative to the same day +5 days (within two weeks) / +40 days (outside two weeks)	BR-12: Telephone verification
Missing materials	Blank (Complete) / Fill in "Examination results, past medical history" (Missing)	Integrity Assessment for BR-01 and BR-02
Clinical decision	accept / reject / request-supplement / refer	BR-04: Prerequisites for Reservation
Funding Source	self-pay / insurance / funding-agency	Approval Determination for BR-14 and BR-15
Consultation Type	simple/finance/clinical(clinical with urgent keyword)	Route determination for BR-30, BR-31, and BR-32

4. Test Scenarios

Each test case follows this format: Test scenario / Preconditions / Test data / Actions / Expected outcomes / Pass–Fail conditions, with annotations specifying the business rules and acceptance criteria it covers. Test cases actually executed in this delivery are marked as **Executed PASS**.

TC-01 End-to-end main path: overall diagram consists of 8 sub-processes in series **Passed**

Test Scenario: Start a patient pathway from the master diagram; eight choices are then sequentially invoked by the Call Activity and completed (Test Scenario).

Prerequisites 3.1 All conditions are met; the SIMULATE switch is not enabled

Test Data: Referral number REF-JOURNEY-1, Appointment number APPT-JOURNEY-1, Case number CASE-JOURNEY-1, Appointment date +5 days; complete registration documents; clinical decision: accept; funding source: self-pay; consultation type: simple

Test Operation Tasklist → "HPAS Overview: Patient Referral — Treatment — Administrative Full Pathway" → Start process; then follow the prompts to complete each sub-process manual task (Actions) in sequence.

Expected outcomes: The invocation order of the sub-processes is P1 → P2 → P3 → P4 → P10 → P7 → P12 → P8; each sub-instance has parentProcessInstanceKey= the overall instance key; all sub-instances and the overall instance are in COMPLETED state (Expected outcomes).

The check is passed if the overall instance state is COMPLETED and the call chain is complete (PASS); otherwise, it fails if any sub-process is missing, the sequence is incorrect, or the overall instance is not completed (FAIL).

coverage BR-01/02/04/10–17/19/23/25/27/30–32、 AC-1、 AC-2、 AC-5

TC-02 Selection 1 · The main path with complete materials **has passed**

Test Scenario Medical secretary registers and refers a patient; all required documents are complete → direct handover to clinical review

Prerequisites: Worker is running; P1 hospital workflow has been deployed.

Test data: the "Materials" field is left blank; Referral source:gp; Attached document: "Referral Form"

Test Operation Complete "Registration Referral" → Wait for the External Worker to automatically execute "Verify Document Completeness"

Expected outcome: The worker writes isComplete=true; the XOR logic follows the "Yes" branch; the task "Confirm handover to clinical review" is triggered; upon completion, a handover notification is sent and the process ends.

Determine whether a handover task has occurred and whether the supplementary item branch has not been taken (PASS); if misjudged as a missing item, then FAIL (BR-02 violation)

coverage BR-01、 BR-02、 AC-1

TC-03 Selection 1 · Missing item replenishment loop (cross-POOL message closed loop) **PASS executed**

Test Scenario Material shortage → Request replacement from external pool (GP / other hospitals) → Receive response from the counterpart → Hospital re-verifies and approves

Prerequisites: The Worker is running; both P1 pools are deployed.

For missing test data, enter "Test Results" in the "Missing Materials" field; for supplementary materials, enter "Test Results and Past Medical History" in the "Supplementary Materials" field.

Test Procedure Complete "Registration Referral" → Wait for the supplementary documentation request message → Complete "Supplementary Materials" for the external pool in the Tasklist → Return to the hospital side for observation

Expected outcomes ① The hospital sets ReferralSupplementRequested and **starts the GP POOL instance**; ② Upon completion, the GP sends ReferralSupplementReceived(association key = referralId); ③ The hospital captures this event, re-verifies it, and when isComplete=true, displays the message "Confirm transfer for clinical review".

The following conditions are evaluated: the GP instance state is COMPLETED AND the hospital instance has progressed to the handover task (PASS); if the follow-up message has not been replied to OR the hospital remains in the waiting state, then FAIL.

coverage BR-01、BR-02、AC-3、AC-4

TC-04 Selection 2 · Accept → Notify the appointment team to schedule an appointment

Test Scenario The consulting physician reviews and "accepts" the request, then sends a cross-Pool notification to the appointment team.

Prerequisites: Worker is running; both P2 pools are deployed

Test data: Decision accept; Rationale: "Meets referral criteria"; Deciding physician: Dr. Chen

Test Operation Complete the "Clinical Review" form → Wait for the Worker to submit a message

Expected outcome: Emit ReferralAccepted; trigger the "Schedule New Patient Appointment" task in the external pool; P2 records the rationale for the decision and the decision-maker.

Pass if the external instance is active and the audit field is non-empty; fail if no message is present or no audit trail is available.

coverage BR-05、BR-06、AC-4、AC-6

TC-05 Selection 2 · Refusal / Referral to another facility → No appointment available

Test Scenario When the decision is to reject or refer, the decision gate in the master diagram must intercept the request.

Prerequisites are the same as those for TC-04; execute the operation to cover the decision gate in the overall layout instance.

Test data Decision reject or refer

Test Operation Complete "Clinical Review" → Observe the overall layout flow

Expected Result Global diagram XOR "Does the referring physician accept?" → follow the "No" branch → terminate at event "Referral not accepted (relevant parties notified)"; **Does not generate** any P3 scheduling instance

If the total number of instances ends with "Not Accepted" and no P3 instances are present (PASS); otherwise, if the system proceeds to scheduling, it results in FAIL (violation of the rule: "No new patient appointments may be scheduled if the case has not been accepted").

coverage BR-04、AC-1、AC-2

TC-06 Selection 3 · Scheduling successful → Confirm appointment → Notify PASS has been executed

Test Scenario Obtain appointment slots via an external scheduling service and confirm the reservation

Prerequisites: Worker is running; P3: both pools are deployed

Test Data: Specialty: Cardiology; Priority: Routine; Expected Time Window; Confirmation Time: 2026-10-01T09:30 Test Workflow: Complete "Create Appointment Request" → Wait for scheduling service acknowledgment → Complete "Confirm Appointment and Write Back"

Expected Result Throw SchedulingRequested → Scheduling Pool instance completes and returns SchedulingResult (scheduleStatus=success) → Hospital takes the success branch → Sends BookingConfirmed via validation Both the hospital and the scheduling instance are in COMPLETED state, and the notification message has been delivered (PASS)

coverage BR-13、BR-04、AC-3、AC-4

TC-07 Selection 3 · Scheduling failed → Remain pending + Timer-based retry

Test Scenario Degradation and retry mechanism for external scheduling services when no available slots are available

Prerequisites Start the Worker with--SIMULATE_SCHEDULING_FAIL_TIMES=2

Test data is the same as TC-06

Test Operation Complete "Create Appointment Request" → Scheduling fails → Complete "Keep as Pending and Record Retries" → Wait for the Timer (PT5S) to automatically return to scheduling

Expected outcome: In case of failure,scheduleStatus=failed and the request remains in Pending state; a "Keep Pending and Record Retries" task is triggered; the scheduled request is automatically resent after the timer expires; the third attempt succeeds;**No duplicate appointments**

If the number of retry attempts and the Timer behavior both meet the expected criteria, the operation ultimately succeeds, and no duplicate source numbers are generated (PASS); otherwise, if the process hangs or duplicate reservations are generated, it fails (FAIL)

coverage BR-10、BR-13、AC-5

TC-08 Option 4 · A phone call must be made within two weeks; if the notification fails, retry PASS executed

Test Scenario Appointment must be scheduled within two weeks → In addition to written correspondence, a phone call must be made and documented.

Precondition Worker is running; two P4 pools are deployed.

Test data Appointment date = current date + 5 days; Notification method: paper; Phone result: connected

Test Procedure Complete "Select notification method" → Wait for the notification worker to send → Complete "Contact by phone and record the result"

Expected result For XOR: If the "Has the appointment been scheduled within two weeks?" check returns "Yes" → initiate a phone contact task; upon completion, record the contact outcome and submit a PatientContactRecorded record;**an external Pool (communication service / patient) instance is activated**

By determination Phone task initiated, contact outcome recorded, external instance started (PASS); however, within two weeks, the task was terminated immediately (FAIL) (BR-12 violation)

Coverage BR-11、BR-12、AC-4、AC-6

TC-09 Selection 4 · Notification send failed → Retry branch (up to 3 attempts)

Test Scenario 重试 and termination of external communication services upon failure

Prerequisites The worker is started with--SIMULATE_NOTIFY_FAIL=1

Test data Appointment date +5 days

Test Operation Complete "Select Notification Method" → Observe the XOR branch

Expected Result: If notifyStatus=false and notifyAttempts<3, return to the "Send Notification" retry step; after reaching 3 attempts, trigger the end event "Notification Send Failed".

If the number of retry attempts is controlled and the process eventually converges (PASS); if there are infinite retry attempts or direct success, then FAIL

coverage BR-11、AC-5

TC-10: Complete the letter within 5–7 days (separation of duties)

Test Scenario Consultant drafts and approves the letter → Medical secretary conducts administrative inspection → Distribution; entire process completed within the specified timeframe

Prerequisites: Worker is running; P10: Both pools are deployed

Test Data Letter type: new-patient; Recipient: GP; Administrative review: passed

Test Procedure Complete the following steps in sequence: "Consultant drafts the letter," "Consultant approves the letter," and "Medical secretary conducts administrative review."

Expected Result: Assign the three tasks to the consultants / medical-secretaries candidate groups; set dispatchedWithin7Days=true for the workers; then complete the process using the "Dispatch on Schedule" option.

Pass if the candidate group is valid and completes within the allotted time; otherwise, fail if the secretary task requires modification of clinical content (BR-25 violation).

coverage BR-23、BR-25、AC-1、AC-6

TC-11 Selection 5 · Unfinished after expiration → Promote to Administrative Manager / Management Level

Test Scenario Letters whose targets have not been met after more than 7 days enter the path monitoring process and trigger escalation.

Prerequisites The Worker is started with--SIMULATE_LETTER_OVERDUE=1(or the distribution timeout is set)

Test Data Letter Type: new-patient

Test Operation Complete three letter tasks → Observe XOR

Expected Result dispatchedWithin7Days=false → Task "Upgrade to Administrative Manager / Management Level" appears → Submitted upon completion
Escalation Raised and **External Pool (Management) Instance Started**

By determining the occurrence of upgrade tasks, the recorded reasons for overdue status, and the initiation of external instances (PASS)

coverage BR-27、AC-4

TC-12 Selection 6 · Funding Subject to Approval (including external institution receipts)

Test Scenario When the insurance/funding agency assumes the costs, register and authorize, then await the external approval acknowledgment.

Prerequisites Worker is running; P7: Both pools are deployed

Test Data Funding Source insurance, Authorization Reference No. AUTH-7788, Application Amount: 1,200

Test Procedure Complete "Determine Funding Source" → Wait for Worker validation → Complete "Approve Funding Application" for the external pool → Return to the hospital to complete "Record Approval"

Expected Result approval Required=true → Throw FundingApproval Requested; the funding agency instance completes the process and returns FundingApproval Result; on the hospital side, the "Record Approval" form is automatically pre-filled with the approval conclusion.

If the external instance COMPLETED is determined and the approval field is populated (PASS), then proceed; otherwise, if no acknowledgment is received, the process fails (FAIL).

coverage BR-14、BR-17、AC-3、AC-4

TC-13 Option 6 · No approval required (self-funded / exempted) → Still requires registration

Test Scenario In self-funded cases, exemptions shall be directly recorded; however, funding information must not be omitted.

Prerequisite The worker is running

Test Data Funding Source self-pay, Amount: 1,200

Test Operation Complete "Determine Funding Source" → Complete "Record Self-Funded / Exempted"

Expected Result approval Required=false → XOR → take the "No" branch → record self-payment / exemption; funding source and amount are then recorded in the audit variables

The following criteria are used for evaluation: if no external approval is required and all fields are complete, then PASS; if any field is missing, then FAIL.

coverage BR-15、BR-14

TC-14 Selection 7 · Payment Successful → Appointment confirmation is then permitted (preliminary gating) **PASS has been executed**

Test Scenario When the patient is required to pay out-of-pocket, the external PSP will collect the payment, and the appointment will be confirmed upon successful payment.

Precondition Worker is running; P12: two pools are deployed

Test data Amount: 1,200; Payment method: card; Card last 4 digits: 4242; Token: tok_1

Test Procedure Complete "Confirm Costs and Payment Method" → Wait for PSP acknowledgment → Complete "Record Payment and Confirm Appointment"

Expected result Trigger PaymentRequested → PSP instance completes and returns PaymentStatusReceived(success + transaction ID) → Hospital follows the success branch → Notifies the treatment appointment team; **Full card number is never processed** (only last four digits / token)

By determination For PSP instances: if the status is COMPLETED, the payment status is successful, and the card number field is absent (PASS); otherwise, if the payment fails, the status is set to FAIL upon confirmation.

Coverage BR-16、BR-17、AC-3、AC-4、AC-6

TC-15 Option 7 · Payment failed / rejected → Investigate + retry or cancel

Test Scenario Failure branch when the issuing bank rejects the request

Prerequisites The Worker is started with--SIMULATE_PAYMENT=failed

Test data: Amount 1200, last four digits of the card: 4242

Test Operation Complete "Confirm Costs and Payment Method" → Complete "Investigate Payment Failure"

Expected Result paymentStatus=failed → Enter the "Failed / Rejected" branch → Record the failure reason and handling approach → Retries / Cancellation

The worker terminates after writing back the processing result;**no duplicate reservations are generated**

Pass if both the failure cause and the handling method are recorded (PASS)

coverage BR-17、AC-5、AC-6

TC-16 Selection 7 · Repeated Deductions / Incomplete Receipts → Abnormal Record

Test Scenario Exception handling when the same charge has already been successfully deducted or when the PSP receipt is incomplete

Prerequisites The Worker is started with--SIMULATE_PAYMENT=duplicate

Test data amount: 1,200

Test Operation Complete "Confirm Costs and Payment Method" → Complete "Record Exception (Duplicate / Incomplete)"

Expected Result paymentStatus=duplicate → Enter the default exception branch → Record the case separately and transfer it to a human operator for follow-up; the process ends at "Exception Recorded"

Determine that an abnormal record exists and no duplicate deductions have occurred (PASS)

coverage BR-19、AC-5

TC-17 Selection 8 · Simple Administrative Inquiry → Direct Call Response

Test Scenario Administrative inquiries shall be responded to on the spot via call handling.

Prerequisites: Worker is running; P8: both pools are deployed

Test data: Inquiry type simple, content: "Confirm appointment time"

Test Operation Complete "Record and Categorize Consultation" → Wait for the automatic categorization Worker → Complete "Direct Response"

Expected outcome: If the XOR flow follows the "Simple Administrative" branch → triggers the "Direct Response" task → records the response content and terminates;**Otherwise**, if the external Pool is not triggered and the validation confirms that the response content has been recorded without exceeding authorization to transfer to clinical services (PASS)

coverage BR-30、BR-32、AC-6

TC-18 Selection 8 · Financial Consulting → Transfer to Financial Pool

Test Scenario Cost-related inquiries must be routed to the finance team; call handlers are prohibited from providing independent explanations.

Prerequisite The worker is running

Test Data Consultation Type finance

Test Operation Complete "Record and Categorize Consultation" → Complete "Financial Processing Consultation" on the financial side → Wait for processing to complete

Expected Result: Throw InquiryRoutedToFinance → The external Pool finance branch instance is started and completed; the process ends once the waiting node is released.

Pass if an external financial instance exists and has been completed

coverage BR-30、AC-4

TC-19 Selection 8 · Clinical Emergency Consultation → Highlight and route to CNS

Test Scenario Clinical consultation (including urgent keywords) routed to the clinical team with highlighted priority handling

Prerequisite The worker is running

Test data: consultation type clinical, urgent keyword "chest pain"

Test Operation Complete "Record and Categorize Consultations" → Complete "Clinical Management Consultation" on the external clinical side

Expected outcome: Throw InquiryRoutedToClinical with the **uRgency field set**; trigger an external clinical branch instance; available handling options: "Return call within the same day / Schedule in-person consultation / Recommend emergency care"

Determine whether urgency persists during message transmission and whether external instances are started (PASS); loss of urgency indicates FAIL (BR-31 violation).

coverage BR-30、BR-31、AC-4

TC-20 Robustness · Invalid data input must not trigger an incident **PASS**

Test Scenario: Error handling when a form is filled with invalid values (e.g., entering 123 for the appointment date)

Prerequisites The worker is running; using the repaired model version

Test Data Appointment Date 123(Non-ISO date)

Test Operation Start the P4 instance and complete "Select Notification Method".

Expected Result: The gateway condition evaluates safely for null/invalid values, and the process proceeds; **the number of incidents is 0** (historical defects are described in Section 7).

If the following conditions are met: IncidenT = 0 and the instance continues (PASS); EXTRACT_VALUE_ERROR occurs, then FAIL

coverage AC-5

TC-21 Structural Compliance · Model deployable and MODeLeR without errors **PASS achieved**

Test Scenarios: Compliance of 9 BPMN structures with the Camunda 8 modeling specifications

Prerequisites `_validate.ps1` is available; the cluster is accessible

Test data: all 9.bpmn files

Run the test operation to perform a structural validation; execute `deploy-all.ps1`; then open the "Problems" pane in Modeler.

Expected Results: unique IDs, no dangling references, DI present for every flow node, valid swimlane references, start and end events present for every process, no isolated nodes, no generic bpmn:task instances, and all message events must include taskDefinition; the Modeler Problems panel must be empty.

The verification 9/9 passed and deployment was successful (PASS)

coverage AC-7

TC-22: Separation of Duties · User Task Candidate Set is Correct

Test Scenarios Candidate groups for clinical, administrative, and financial tasks must not overlap.

Prerequisites Worker is running; all processes are deployed

test data —

Test Operation Launch each workflow individually and check the candidate groups for each task in the Tasklist.

Expected Roles Clinical =consultants/clinical-team; Administrative =medical-secretaries/appointment team/call-handling;

Financial =finance-team; External Parties =gp-practices/funder reviewers, etc.

By determining that the non-administrative group holds clinical decision-making tasks, and the clinical group holds financial refund tasks (PASS)

coverage BR-01、BR-25、AC-6

TC-23 Traceability · Key actions are properly logged, and message association keys are correct

Test scenarios— including determination, contact, payment, funding, and correspondence — can all be traced back to individuals and specific time points.

Prerequisites: At least execute TC-03, TC-08, TC-12, and TC-14

Test Data Data for each test case

Test Operation Retrieve instance variables in Operate

Expected Results include reviewForm.decidedBy,checkNote,contactResult,pspReference,approvalRecordForm.*, and each *At timestamp; message association key = business number (referralId / appointmentId / caseId)

By verifying that the audit fields are complete and the association keys can be traced back to the patient pathway (PASS)

coverage AC-6

5. Test Execution Log has been executed

The following are the test cases that were actually executed and passed on the local Camunda 8 cluster during this delivery (and are reproducible).

Use case	Execution Content	Key Evidence (Process Example / Results)
TC-03	P1 Component Replenishment Loop (Dual-Pool Message Closed Loop)	Hospital instance 2251799813731802; GP instance 2251799813732423= COMPLETED; Hospital has progressed to "Confirm Handover Clinical Review".
TC-06	P3 Scheduling Request / Result	Hospital instance 2251799813739915= COMPLETED; Scheduling instance 2251799813740046= COMPLETED
TC-14	P12 Payment Request / Acknowledgment	PSP instance 2251799813739569= COMPLETED; The hospital has successfully completed its branch operations.
TC-01	End-to-end main path (8 sub-processes in series)	Total instances 2251799813750025= COMPLETED; P1/P2/P3/P4/P10/P7/P12/P8 are all COMPLETED; 16 manual tasks; incident = 0
TC-01	Start an external instance via cross-Pool messaging	P2_PartnerPool 2251799813750238 、 P3_SchedulingPool 2251799813750311 (COMPLETED) 、 P4_PatientPool 2251799813750582 、 P12_PSPPool 2251799813751031 (COMPLETED)
TC-20	Dirty data (appointment date 123) re-measurement	After the fix, incident = 0 , and the workflow proceeds (pre-fix state was EXTRACT_VALUE_ERROR,see Section 7)
TC-21	Structure Validation + One-Click Deployment	September 9: Model structure validation passed; 48 files deployed successfully in a single deployment run (17 process definitions + 39 form versions)

6. Coverage Matrix (BR ↔ Test Cases)

Business Rules	Main points	Covered Use Cases
BR-01	The medical secretary is solely responsible for administrative registration and does not conduct clinical assessments.	TC-02、 TC-03、 TC-22
BR-02	The case will be handed over to the consulting physician only after all documentation has been complete.	TC-02、 TC-03
BR-04	Appointments for new patients will be scheduled only for accepted referrals.	TC-05、 TC-06

BR-05 / BR-06	The rationale for the decision and the identity of the decision-maker must be duly documented.	TC-04、 TC-23
BR-10	If scheduling fails, keep the status as "Pending" and record a retry attempt.	TC-07
BR-11	Select notification channels based on patient preferences	TC-08、 TC-09
BR-12	A telephone contact must be made within two weeks, and the results must be documented.	TC-08
BR-13	Once the scheduling is successful, confirm it and reply.	TC-06、 TC-07
BR-14 / BR-15	Registration of Funding Sources and Amounts (including cases exempt from approval)	TC-12、 TC-13
BR-16	Confirm the amount due and the payment method; do not store the full card number.	TC-14
BR-17	Reasons for Payment Failure Registration and Corresponding Handling Procedures	TC-12、 TC-15
BR-19	Repeated/Incomplete Payment: Separate Record	TC-16
BR-23	The consultant drafts and approves the letter.	TC-10
BR-25	The term "secretary" refers solely to administrative duties and does not carry a clinical connotation.	TC-10、 TC-22
BR-27	Upgrade to Administrative Manager / Management Level in over 7 days	TC-11
BR-30	Consultation registration, classification, and priority determination	TC-17、 TC-18、 TC-19
BR-31	Urgent keywords are conveyed along with the message and highlighted.	TC-19
BR-32	Direct response to simple administrative inquiries	TC-17
OQ-01	Cross-Pool processing time limit (to be agreed upon with operations)	TC-11 (temporarily implemented under this plan following overdue upgrade)

Acceptance criteria cover: AC-1 ← TC-01/02/05/06/10; AC-2 ← TC-01/05; AC-3 ← TC-03/06/12; AC-4 ← TC-03/04/08/11/14/18/19; AC-5 ← TC-07/09/15/16/20; AC-6 ← TC-04/08/10/12/14/17/22/23; AC-7 ← TC-21.

7. Defect Log

Number	Phenomenon	Root cause	Processing and Re-measurement
D-01	Select 4 in "Is the appointment within two weeks?" to trigger an incident:EXTRACT_VALUE_ERROR,error message:Failed to parse date from '123'	The gateway condition directly calls date (appointmentDate); invalid input returns null, and the XOR operation fails to yield a boolean value; furthermore, no format validation is applied to the form.	The condition is modified to skip string comparison using date() and implement null protection; the form is now equipped with YYYY-MM-DD regular expression validation. Re-run the test with the same invalid input: incident = 0 (TC-20 PASS)
D-02	The modeling process for the external pool in Modeler reports Process is missing start event	The external pool is for isExecutable= "false": Zeebe ignores this during deployment (deployment succeeds), but Modeler does not differentiate based on executability during validation.	Complete the external Pool with standardized start and end events, as well as full-connect sequential flows; additionally, replace the generic bpmn:task with the native Camunda 8 userTask.

			/serviceTask. Re-run of Modeler succeeded without errors (TC-21 PASS)
D-03	Select 1: The model contains an isolated redundant node Activity_1 kuvs54(no name, no connections)	Redundant elements in the modeling process; Camunda 8 does not support the generic bpmn:task	Combined with the lane references, these were cleared along with DI; after adding the rule "No use of generic tasks / No isolated nodes" to the structural validation, re-testing yielded a pass rate of 9/9.

8. Pass–Fail Criteria

- **Test case passed:** all expected outcomes of this test case were met, and **incident = 0** during execution (AC-5).
- **Test case failure:** none of the expected outcomes occur, the workflow does not reach the designated end event, an incident is triggered, or a business rule violation occurs (e.g., scheduling an appointment without approval, failing to make a phone call within two weeks, or confirming an appointment before payment is successful).
- **Overall acceptance pass:** All cases from TC-01 to TC-23 passed, and each case from AC-1 to AC-7 was supported by at least one test case; the coverage matrix for BR-01 to BR-32 contained no gaps.
- **Defect Severity Classification:** *Critical*= Violation of BR or triggering an incident (must be fixed before re-testing); *Minor*= Partial loss of expected output, but the workflow can proceed; *Trivial*= Copywriting/layout issues.
- **Re-testing Requirements:** After fixing any critical defect, the original test cases must be re-run using **the same test data**, and a new regression test case record must be added.